BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

MATHS PAPER- 9th — ANNUAL EXAM-2021 — SECTION-A)

Time Allowed: 20mints | **Maximum Marks:** 15

Note: Q. 1: Choose the correct options i.e. A, B, C, or D.

Q1. The matrix [-3 0; 0 -3] is ..... matrix.

(A) Diagonal
(B) Scalar
(C) Additive identity
(D) None

Q2. {0, 1, 2, 3, 4.....} set of .....numbers.

(A) Natural
(B) Whole
(C) Irrational
(D) Integers.

Q3. Characteristic of log (3958) is.....

(A) 4
(B) 3
(C) - 3
(D) - 4

Q4. If $\log_6 x = 3$, then x = ...

(A) 36
(B) 84
(C) 216
(D) 221

Q5. Degree of polynomial $P(t) = 5t^2 - 10t + 4$ is .....

(A) 1
(B) 2
(C) 3
(D) 0

Q6. Factor of 2x - 8 are .....

(A) 2
(B) x - 4
(C) 2, x - 4
(D) x + 4

Q7. L.C.M of $(a - b)^4$ and $(a - b)^3$ is.....

(A) $(a - b)$ (B) $(a - b)^2$

(C) $(a - b)^4$ (D) $(a - b)^7$

Q8. The solution set of $\sqrt{x} = -10$ are.....

(A) {100} (B) {10}

(C) {-10} (D) {}

Q9. The point (4, 6) is located in a ..... quadrant.

(A) 1st (B) 2nd

(C) 3rd (D) 4th

Q10. Two perpendicular segments meet in a .....angle.

(A) 45° (B) 60°

(C) 90° (D) 180°

Q11. If sum of measure of < A and < C of a parallelogram ABCD is 130°, then < B = ...

(A) 25° (B) 50°

(C) 65° (D) None of them

Q12. In a quadrilateral triangle, all the perpendicular bisects are....

(A) Congruent (B) Concurrent

(C) The angle bisector as well (D) All of these

Q13. ..... Represented sides of a right angled triangle.

(A) 2,3,5 (B) 3,4,5

(C) 4,5,6 (D) 5,6,7

Q14. Perpendicular from the vertex of a triangle to its opposite side is called .....

(A) Median (B) Perpendicular bisector

(C) Altitude (D) angle bisector

Q15. In practical geometry, it is necessary to write.....

(A) Given (B) To prove

(C) Step of construction (D) Proof

SECTION-B

Time Allowed: 2.40 hrs. | **Maximum Marks:** 36

Q2. (Marks 36) Q2: Answer any Nine 9 parts. Each part carries 4 marks.

1. Find the additive inverse of the B = [a -a b; -c a -b; l m n]

2. If A = [0 -1; 2 1], B = [2 3; 1 0], then show that $(AB)^{-1} = B^{-1} A^{-1}$

3. Simplify: (2^(p+1) 3^(2p-q) 5^(p+q) 6^q) / (6^p 10^(q+2) 15^q)

4. Simplify $2.83 / (6.52)^2$ with help of logarithm

5. Factorize $1 + 2UV – U^2 – V^2$

6. Factorize $8x^2 – 4x + 2/(3x) – 1/(27x^3)$

7. Find L.C.M of $x^3 - 6x^2 + 11x – 6$ and $x^3 – 4x + 3$

8. What should be added to the expression $9x^4 + 12x^3 + 10x^2 – 3x – 3$ to make a perfect square.

9. Age of a mother is 13 times the age of her daughter. It will be only five times after four years. Find their present ages.

10. Solve the radical equation $3\sqrt{x} + 7 = 10$

11. Graph the equation y = 6

12. When a = 2, b = - 3, and c = 3, evaluate 2a – 3b + 2c.

SECTION-C

Maximum Marks: 24

> Note: Q3: Attempt any three (3) questions. All carries equal marks.

Q2. Prove that the point A (1, 2), B (0, 4) and C (3, 5) are the vertex of a scalene Triangle?

Q3. Prove that any point on the right bisector of a line segment is equidistant from its end point.

Q4. The sum of the lengths of any two sides of triangle is greater than the length of the third sides.

Q5. Construct △ABC, draw their altitude and verify their concurrency. m PQ = 6 cm, when m AB= 5.4 cm, m ABC= 6 cm and m CA = 5.4cm. Also draw their angle bisectors